Allocation Model Draft

James Evans & Chong Han

07/28/2025

James:

Risk Free Asset: Cox–Ingersoll–Ross (CIR) Model

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dZ_t$$

- r_t: The short-term interest rate at time t.
- κ: Speed of mean reversion — how quickly r_t moves toward the long-run mean.
- θ: Long-run mean level of the short term rate.
- σ: Volatility parameter; higher σ means more random fluctuations. It is strictly positive.
- $\sqrt{r_t}$: State-dependent volatility; ensures non-negative rates.
- dZ_t: Represents the stochastic shock over the infinitesimal interval $[t, t + dt]$.

Sign Analysis of Terms in the CIR Model

Term	Description	Possible Values
r_t	Short-term interest rate	≥ 0
θ	Mean short-term rate	> 0
κ	Mean reversion speed	> 0
$\theta - r_t$	Distance from long-run mean	> 0, $= 0$, or < 0
dt	Infinitesimal time increment	> 0
$\kappa(\theta - r_t)$	Deterministic mean-reversion force	> 0, $= 0$, or < 0
$\kappa(\theta - r_t)dt$	Drift term	> 0, $= 0$, or < 0
$\sqrt{r_t}$	Volatility scale (CIR-specific)	≥ 0, zero when $r_t = 0$
σ	Volatility of volatility	> 0
dZ_t	Standard Brownian motion increment	> 0, $= 0$, or < 0
$\sigma\sqrt{r_t}$	Volatility coefficient	≥ 0
$\sigma\sqrt{r_t}dZ_t$	Diffusion term	> 0, $= 0$, or < 0
dr_t	Interest rate differential	> 0, $= 0$, or < 0

State-dependent volatility

The randomness is proportional to $\sqrt{r_t}$, which **grows when r_t is large**, and **shrinks when r_t is small**. This is more realistic than constant volatility because **interest rates don't fluctuate wildly when they're near zero**.

Keeps $r_t \geq 0$ (non-negative interest rates)

A natural question is: what prevents r_t from becoming negative, especially since the square root $\sqrt{r_t}$ becomes undefined for $r_t < 0$? The answer lies in the structure of the diffusion and drift terms. As $r_t \to 0$, the volatility $\sigma\sqrt{r_t}$ also tends to zero, meaning that random fluctuations vanish near the boundary. At the same time, the drift term becomes approximately $\kappa\theta\, dt$, which is strictly positive and pushes the process upward. Thus, the process becomes increasingly deterministic and mean-reverting as it approaches zero. This dynamic causes the process to "bounce off" the boundary rather than cross it. A formal guarantee of non-negativity is provided by the *Feller condition*, which states that if $2\kappa\theta \geq \sigma^2$, then the boundary at $r_t = 0$ is inaccessible and the process remains strictly non-negative almost surely. If the Feller condition is violated, the process may still stay non-negative, but can touch zero; in severe cases, it may cross below zero, at which point the square root becomes undefined and the SDE breaks down. Therefore, both the vanishing of the volatility near zero and the strength of the mean-reverting drift explain why the CIR model is able to maintain $r_t \geq 0$.

Dimensional Analysis of the CIR Model

Consider the Cox–Ingersoll–Ross (CIR) model for the short-term interest rate:

$$dr_t = \kappa(\theta - r_t)\, dt + \sigma\sqrt{r_t}\, dZ_t$$

What Are the Units?

At first glance, one might think that r_t, being a "rate," should have units of $\frac{1}{\text{time}}$, and that κ is a unitless scalar. However, this is not the case in finance. Let's break it down carefully.

Units of r_t

The interest rate r_t is a **dimensionless scalar** — it represents a proportion (e.g., 5% interest means $r_t = 0.05$).

$$[r_t] = 1$$

Left-Hand Side — dr_t

Since r_t is dimensionless, its infinitesimal change dr_t must also be dimensionless:

$$[dr_t] = 1$$

Drift Term — $\kappa(\theta - r_t)\, dt$

We now examine the units of each piece:

- $\theta - r_t$: dimensionless
- dt: units of time

To ensure dimensional consistency with dr_t, we need:

$$[\kappa] = \frac{1}{\text{time}} \quad \Rightarrow \quad [\kappa(\theta - r_t)dt] = \frac{1}{\text{time}} \cdot 1 \cdot \text{time} = 1$$

So the drift term is dimensionless, just like dr_t.

Diffusion Term — $\sigma\sqrt{r_t}\,dZ_t$

Now for the stochastic component:

- $\sqrt{r_t}$: still dimensionless, since r_t is dimensionless
- dZ_t: Brownian motion increment, which has units of $\sqrt{\text{time}}$

To cancel the time units and ensure this term is dimensionless, we require:

$$[\sigma] = \frac{1}{\sqrt{\text{time}}} \quad \Rightarrow \quad [\sigma\sqrt{r_t}\,dZ_t] = \frac{1}{\sqrt{\text{time}}} \cdot 1 \cdot \sqrt{\text{time}} = 1$$

Risky Asset: Heston Stochastic Volatility Model Equation One

$$dP_1(t) = \mu P_1(t)\,dt + \sqrt{v_t}P_1(t)\,dZ_t$$

- $P_1(t)$: Price of the risky asset at time t.
- μ: Expected return on the asset (1 / unit of time).
- $\sqrt{v_t}$: Stochastic volatility — depends on variance process v_t.
- dZ_t: Represents the stochastic shock over the infinitesimal interval $[t, t + dt]$.

Units of Brownian Motion W_t

By definition:

$$\text{Var}(Z_t) = \mathbb{E}\left[(Z_t - \mathbb{E}[Z_t])^2\right]$$

For a standard Brownian motion, we know:

$$\mathbb{E}[Z_t] = 0 \quad \Rightarrow \quad \text{Var}(Z_t) = \mathbb{E}[Z_t^2]$$

Also, from Brownian motion properties:

$$Z_t \sim \mathcal{N}(0, t) \quad \Rightarrow \quad \text{Var}(Z_t) = t$$

So putting it together:

$$\mathbb{E}[Z_t^2] = t$$

Assume time t is measured in seconds. Then:

$$[\mathbb{E}[Z_t^2]] = \text{seconds} \quad \Rightarrow \quad [Z_t^2] = \text{seconds} \quad \Rightarrow \quad [Z_t] = \sqrt{\text{seconds}}$$

In the Heston stochastic differential equation:

$$dP_1(t) = \mu P_1(t)\,dt + \sqrt{v_t}P_1(t)\,dZ_t$$

We require both terms on the right-hand side to have units of dollars.

Drift term

$[\mu] = 1/\text{time}, \quad [dt] = \text{time} \Rightarrow [\mu P_1(t)\,dt] = \text{dollars}$

Diffusion term

$[v_t] = 1/\text{time} \Rightarrow [\sqrt{v_t}] = 1/\sqrt{\text{time}}$

$[dZ_t] = \sqrt{\text{time}}$

So $[\sqrt{v_t} \cdot dZ_t] = 1 \Rightarrow [\sqrt{v_t}P_0(t)\,dZ_t] = \text{dollars}$

Risky Asset: Heston Stochastic Volatility Model Equation Two

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dZ_t$$

- v_t: Stochastic variance (i.e., $v_t = \sigma_t^2$).
- κ: Speed of mean reversion for variance.
- θ: Long-run mean of variance.
- σ: Volatility of variance — controls how volatile v_t is.
- dZ_t: Represents the stochastic shock over the infinitesimal interval $[t, t+dt]$.
- $\sqrt{v_t}$: Ensures variance stays non-negative.

Dimensional Analysis of the Heston Variance Process

Consider the stochastic variance equation in the Heston model:

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dZ_t$$

Units of v_t

The process v_t represents the *instantaneous variance* of the asset return process, meaning:

$$v_t = \sigma_t^2 \quad \Rightarrow \quad [v_t] = \left(\frac{1}{\sqrt{\text{time}}}\right)^2 = \frac{1}{\text{time}}$$

Hence:

$$[dv_t] = \frac{1}{\text{time}}$$

Drift Term: $\kappa(\theta - v_t)\,dt$

- θ and v_t both have units of 1/time
- $\theta - v_t$: has units of 1/time
- dt: has units of time

To make the product dimensionally consistent with dv_t, we require:

$$[\kappa] = \frac{1}{\text{time}}$$

Then:

$$[\kappa(\theta - v_t)dt] = \frac{1}{\text{time}} \cdot \frac{1}{\text{time}} \cdot \text{time} = \frac{1}{\text{time}}$$

Diffusion Term: $\sigma\sqrt{v_t}\,dZ_t$

- $\sqrt{v_t}$: has units $1/\sqrt{\text{time}}$
- dZ_t: has units $\sqrt{\text{time}}$

To ensure the entire term has units 1/time, we require:

$$[\sigma] = \frac{1}{\text{time}}$$

Then:

$$[\sigma\sqrt{v_t}\,dZ_t] = \frac{1}{\text{time}} \cdot \frac{1}{\sqrt{\text{time}}} \cdot \sqrt{\text{time}} = \frac{1}{\text{time}}$$

Wealth Process with Consumption Term

$$dW^S(t) = S(t) \cdot \frac{dP_1(t)}{P_1(t)} + (W(t) - S(t)) \cdot \frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt$$

- $S(t)\frac{dP_1(t)}{P_1(t)}$: Return from wealth invested in the risky asset.
- $(W(t) - S(t))\frac{dP_0(t)}{P_0(t)}$: Return from the wealth invested in the risk-free asset.
- $-(\vartheta + \beta - d)S(t)\,dt$: Adjustment for transaction costs, taxes, and dividends.
- $-C(t)\,dt$: Rate of consumption, which directly reduces wealth over time.
- $C(t)$: Control variable representing the investor's consumption rate at time t.

Note: We Assume consumption is a fixed proportion of wealth:

$$C(t) = \kappa W(t), \quad \kappa \in (0, 1)$$

Plugging in CIR and Heston into the Wealth Equation

First, we substitute the Heston model for the risky asset

$$\frac{dP_1(t)}{P_1(t)} = \mu\,dt + \sqrt{v_t}\,dZ_t.$$

Next, because the price process of the risk-free asset satisfies the ordinary differential equation,

$$dP_0(t) = r_t P_0(t)\,dt,$$

we substitute

$$\frac{dP_0(t)}{P_0(t)} = r_t\,dt$$

for the risk-free asset.

Note: Here, the process r_t is used symbolically to define the return on the risk-free asset. We do not need a closed-form solution for r_t; it is treated as a state variable whose evolution is governed by the CIR SDE.

Substituted Wealth Process with Consumption Term

$$\begin{aligned}
dW^S(t) &= S(t) \cdot (\mu\,dt + \sqrt{v_t}\,dZ_t) + (W(t) - S(t)) \cdot r_t\,dt - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt \\
&= \mu S(t)\,dt + \sqrt{v_t}S(t)\,dZ_t + W(t)r_t\,dt - S(t)r_t\,dt - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt \\
&= [\mu S(t) + W(t)r_t - S(t)r_t - (\vartheta + \beta - d)S(t) - C(t)]\,dt + \sqrt{v_t}S(t)\,dZ_t \\
&= [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)]\,dt + \sqrt{v_t}S(t)\,dZ_t
\end{aligned}$$

Deriving the Hamilton-Jacobi-Bellman Equation

We define the value function $J(t, r(t), W(t))$, where:

- t is time,
- $r(t)$ is the short rate process,
- $W(t)$ is the wealth process,
- $S(t)$ is the control (investment in risky asset).

Itô's Lemma (Multivariate Form)

Let $J(t, r(t), W(t))$, where both $r(t)$ and $W(t)$ are Itô processes of the form $dX_t = \mu_t\, dt + \sigma_t\, dZ(t)$, driven by the same Brownian motion $Z(t)$. Then:

$$dJ = J_t\, dt + J_r\, dr + J_w\, dW + \frac{1}{2} J_{rr}\, (dr)^2 + \frac{1}{2} J_{ww}\, (dW)^2 + J_{rw}\, dr\, dW$$

Computing Quadratic Variation Terms

We apply Itô's Lemma to the value function $J(t, r_t, W_t)$, where both r_t and W_t are Itô processes driven by the same Brownian motion Z_t.

We use the fundamental stochastic calculus identities:

$$(dZ_t)^2 = dt, \quad dt \cdot dZ_t = 0, \quad (dt)^2 = 0$$

Short Rate Process

Recall the Short Rate Process (Cox–Ingersoll–Ross):

$$dr_t = \kappa(\theta - r_t)\, dt + \sigma\sqrt{r_t}\, dZ_t.$$

We compute the second-order differential $(dr_t)^2$ as follows:

$$\begin{aligned}
(dr_t)^2 &= [\kappa(\theta - r_t)\, dt + \sigma\sqrt{r_t}\, dZ_t]^2 \\
&= \kappa^2(\theta - r_t)^2(dt)^2 + 2\kappa(\theta - r_t)\sigma\sqrt{r_t}\, dt\, dZ_t + \sigma^2 r_t (dZ_t)^2 \\
&= 0 + 0 + \sigma^2 r_t\, dt \\
&= \sigma^2 r_t\, dt
\end{aligned}$$

Wealth Process

The wealth process is given by:

$$dW^S(t) = [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)]\, dt + \sqrt{v_t} S(t)\, dZ_t$$

Define:

$$A(t) := \mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)$$

Then the wealth process becomes:

$$dW^S(t) = A(t)\, dt + \sqrt{v_t} S(t)\, dZ_t$$

Now compute the square of the differential:

$$\begin{aligned}
(dW^S(t))^2 &= [A(t)\, dt + \sqrt{v_t} S(t)\, dZ_t]^2 \\
&= A(t)^2 (dt)^2 + 2A(t)\sqrt{v_t} S(t)\, dt\, dZ_t + v_t S(t)^2 (dZ_t)^2 \\
&= 0 + 0 + v_t S(t)^2\, dt \\
&= v_t S(t)^2\, dt
\end{aligned}$$

Cross Term

We compute the cross-variation term between the short rate process and the wealth process:

$$\begin{aligned} dr_t \cdot dW^S(t) &= [\kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dZ_t] \cdot [A(t)\,dt + \sqrt{v_t}S(t)\,dZ_t] \\ &= \kappa(\theta - r_t)A(t)\,(dt)^2 + \kappa(\theta - r_t)\sqrt{v_t}S(t)\,dt\,dZ_t + \sigma\sqrt{r_t}A(t)\,dZ_t\,dt + \sigma\sqrt{r_t}\sqrt{v_t}S(t)\,(dZ_t)^2 \\ &= 0 + 0 + 0 + \sigma\sqrt{r_t v_t}\,S(t)\,dt \\ &= \sigma\sqrt{r_t v_t}\,S(t)\,dt \end{aligned}$$

These second-order terms are required when applying Itô's Lemma to $J(t, r_t, W_t)$, and appear in the final HJB equation through the terms:

$$\frac{1}{2}J_{rr}(dr_t)^2, \quad \frac{1}{2}J_{ww}(dW_t)^2, \quad J_{rw}\,dr_t \cdot dW_t$$

Applying Itô's Lemma (Multivariate Form)

$$\begin{aligned} dJ = J_t\,dt &+ J_r\,[\kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dZ_t] \\ &+ J_w\,\left[(\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t))\;dt + \sqrt{v_t}S(t)\,dZ_t\right] \\ &+ \frac{1}{2}J_{rr}\,\sigma^2 r_t\,dt + \frac{1}{2}J_{ww}\,v_t S(t)^2\,dt + J_{rw}\,\sigma\sqrt{r_t v_t}\,S(t)\,dt \end{aligned}$$

Note: Within this context, $dr_t = d_r$.

Extract Drift Terms for the HJB

The HJB uses only drift terms (the dt-terms). Group them:

$$\begin{aligned} 0 = J_t &+ \kappa(\theta - r_t)\,J_r \\ &+ [\mu S(t) + r_t(W(t) - S(t)) - (\vartheta + \beta - d)S(t) - C(t)]\,J_w \\ &+ \frac{1}{2}\sigma^2 r_t\,J_{rr} + \frac{1}{2}v_t S(t)^2\,J_{ww} + \sigma\sqrt{r_t v_t}\,S(t)\,J_{rw} \end{aligned}$$

This is the Hamilton-Jacobi-Bellman (HJB) equation corresponding to the wealth and short rate dynamics under control $S(t)$.

Finding the Optimal Control

Taking the derivative with respect to $S(t)$, we get:

- $\mu S(t) J_w \quad \Rightarrow \quad \mu J_w$
- $-r_t S(t) J_w \quad \Rightarrow \quad -r_t J_w$
- $-(\vartheta + \beta - d)S(t)J_w \quad \Rightarrow \quad -(\vartheta + \beta - d)J_w$
- $\frac{1}{2}v_t S(t)^2 J_{ww} \quad \Rightarrow \quad v_t S(t) J_{ww}$
- $\sigma\sqrt{r_t v_t}S(t)J_{rw} \quad \Rightarrow \quad \sigma\sqrt{r_t v_t}J_{rw}$

Combining all terms:

$$\frac{dF}{dS(t)} = (\mu - r_t - \vartheta - \beta + d)J_w + v_t S(t) J_{ww} + \sigma\sqrt{r_t v_t}J_{rw}$$

$$\text{Set } \frac{dF}{dS(t)} = 0 : \quad [\mu + d - r_t - (\vartheta + \beta)]\,J_w + \sigma\sqrt{r_t v_t}J_{rw} + v_t S(t) J_{ww} = 0$$

Solve for the optimal control $S^*(t)$:

$$S^*(t) = -\frac{[(\mu + d) - (r_t + \vartheta + \beta)]J_w + \sigma\sqrt{r_t v_t}J_{rw}}{v_t J_{ww}} \tag{34}$$

Applying the CRRA Ansatz to Eliminate Wealth Dependency

Eliminating Dependency on w

To eliminate dependency on w, we conjecture that

$$J(t,r,w) = G(t,r) \cdot \frac{w^{1-\phi}}{1-\phi}, \quad \phi \neq 1 \tag{35a}$$

with the boundary condition

$$G(T,r) = 1.$$

Note: Since $r(t) = r$ and $W(t) = w$, we can write:

$$J(t, r(t), W(t)) = J(t,r,w)$$

From (35a), we obtain:

$$J_w = w^{-\phi} G, \quad J_{ww} = -\phi w^{-\phi-1} G, \quad \text{and} \quad J_{rw} = w^{-\phi} G_r \tag{35c}$$

Substituting (35c) into the Optimal Control Formula

We begin with the optimal control expression from the new equation:

$$S^*(t) = -\frac{[(\mu+d) - (r_t + \vartheta + \beta)] J_w + \sigma\sqrt{r_t v_t} J_{rw}}{v_t J_{ww}}$$

From equation (35c), we know the following expressions for the derivatives of $J(t,r,w)$:

$$\begin{aligned} J_w &= w^{-\phi} G, \\ J_{ww} &= -\phi w^{-\phi-1} G, \\ J_{rw} &= w^{-\phi} G_r \end{aligned}$$

Substitute these into the expression for $S^*(t)$:

$$\begin{aligned} S^*(t) &= -\frac{[(\mu+d) - (r_t + \vartheta + \beta)] \cdot w^{-\phi} G + \sigma\sqrt{r_t v_t} \cdot w^{-\phi} G_r}{v_t \cdot (-\phi w^{-\phi-1} G)} \\ &= \frac{[(\mu+d) - (r_t + \vartheta + \beta)] \cdot w^{-\phi} G + \sigma\sqrt{r_t v_t} \cdot w^{-\phi} G_r}{\phi v_t w^{-\phi-1} G} \end{aligned}$$

Now simplify numerator and denominator:

$$S^*(t) = \frac{w}{\phi v_t} \left[(\mu+d) - (r_t + \vartheta + \beta) + \sigma\sqrt{r_t v_t} \cdot \frac{G_r}{G} \right] \tag{36}$$

This is the final closed-form expression for the optimal strategy $S^*(t)$ in terms of current wealth w, short rate r_t, and the reduced value function $G(t,r)$.

Eliminating Dependency on r

Further to eliminate dependency on r we conjecture that

$$G(t,r) = h(t) \left[\frac{r^{1-\phi}}{1-\phi} \right], \tag{37a}$$

such that

$$h(T) = \frac{1-\phi}{r^{1-\phi}}. \tag{37b}$$

We obtain from (37a)

$$G_r = r^{-\phi} h(t). \tag{37c}$$

Substituting (37a) and (37c) into (36)

From equation (36), we have:

$$S^*(t) = \frac{w}{\phi v_t}\left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma\sqrt{r_t v_t}\cdot\frac{G_r}{G}\right]. \tag{36}$$

Using the ansatz from (37a)

$$G(t,r) = h(t)\left[\frac{r^{1-\phi}}{1-\phi}\right], \tag{37a}$$

and its derivative,

$$G_r = r^{-\phi}h(t), \tag{37c}$$

we compute

$$\frac{G_r}{G} = \frac{r^{-\phi}h(t)}{h(t)\cdot\frac{r^{1-\phi}}{1-\phi}} = \frac{(1-\phi)r^{-\phi}}{r^{1-\phi}} = \frac{(1-\phi)}{r}.$$

Substitute this into (36):

$$\begin{aligned} S^*(t) &= \frac{w}{\phi v_t}\left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma\sqrt{r_t v_t}\cdot\frac{1-\phi}{r_t}\right] \\ &= \frac{w}{\phi v_t}\left[(\mu + d) - (r_t + \vartheta + \beta) + \sigma\sqrt{v_t}\cdot\frac{1-\phi}{\sqrt{r_t}}\right]. \end{aligned}$$

This expression eliminates dependency on G, leaving $S^*(t)$ in terms of w (the wealth of the investor at time t), v_t, and r_t only.

Long-Run Behavior of the Optimal Strategy

As $t \to \infty$, the state variables r_t and v_t, which follow mean-reverting CIR processes, converge to their respective long-run means:

$$r_t \to \theta_r, \quad v_t \to \theta_v.$$

Substituting these asymptotic values into the expression for $S^*(t)$, we obtain:

$$\lim_{t\to\infty} S^*(t) = \frac{w}{\phi\theta_v}\left[(\mu + d) - (\theta_r + \vartheta + \beta) + \sigma\sqrt{\theta_v}\cdot\frac{1-\phi}{\sqrt{\theta_r}}\right].$$

This substitution is justified because in the long run, the transient fluctuations in r_t and v_t average out, and their stationary distributions concentrate around their respective means. Thus, the long-run optimal control depends only on model parameters and wealth w, reflecting a stabilized investment strategy.

Chong:

TO BE CONT. Contents Still Needed Modifications

1 Notes on the Cumulative Net Contribution Benchmark in Continuous-Time Optimal Asset Allocation

1.1 Introduction and Motivation

In our research, we explore how **regret aversion** shapes investors' optimal portfolio choices. Specifically, we model regret aversion relative to a benchmark defined by the **cumulative net contributions** an investor has made over time. Unlike traditional benchmarks like market indices (e.g., S&P 500), our cumulative net contribution benchmark captures personalized investor expectations, building on regret-based models like those of Bell (1982) or Loomes and Sugden (1982).Referred to the outline.

Before diving into the mathematical details, let's clearly define this benchmark conceptually:

- **Cumulative Net Contribution Benchmark (M_t)**:
  - Represents the total amount of money an investor has cumulatively deposited (or contributed) into their investment portfolio, minus any withdrawals, from time 0 up to time t.
  - It is a **reference point** against which investors measure success or failure.
  - Intuitively, if investors find their current wealth falls short of their total contributions, they experience regret—reflecting dissatisfaction about their investment outcomes.

Why this matters in our research:

- In reality, investors often gauge success not only by absolute returns but by comparing outcomes against the total money they put into the market.
- Capturing regret aversion through this benchmark makes our model behaviorally realistic, reflecting real-world investor psychology, especially relevant in contexts like pension investing and lifecycle wealth management.

1.2 Mathematical Definition of the Benchmark

I proposed a **Dynamic Performance-Based Contribution Benchmark**:

Proposed Expression:

$$\boxed{dM_t = \left[c_0 + \eta(W_t - M_t)^+ - \xi(M_t - W_t)^+\right] dt, \quad M_0 = 0}$$

where we use the notation:

- $x^+ = \max(x, 0)$, indicating we only consider positive deviations in each direction separately.

Parameter Definitions:

- M_t: Cumulative net contributions (benchmark) at time t.
- dM_t: Incremental change in cumulative contributions over the infinitesimal interval $[t, t + dt]$.
- W_t: Current wealth at time t.
- $c_0 \geq 0$: Constant, baseline contribution rate ensuring continuous saving.
- $\eta \geq 0$: Sensitivity parameter—how strongly investors **increase contributions** when their wealth exceeds their benchmark (optimism-driven behavior).

- $\xi \geq 0$: Sensitivity parameter—how strongly investors **decrease contributions** when wealth falls below their benchmark (pessimism-driven behavior).

Economic Interpretation of the Expression:

- If wealth exceeds cumulative contributions ($W_t > M_t$): the investor feels successful, increasing future contributions at a rate proportional to the surplus, capturing optimism or positive feedback.
- If cumulative contributions exceed wealth ($M_t > W_t$): the investor feels behind or unsuccessful, thus reducing contributions proportionally, capturing pessimism or negative feedback.

Parameter Estimation:

In practice, parameters η and ξ can be calibrated using investor behavior data, such as pension contribution patterns or survey-based measures of regret sensitivity. For example, η might be estimated by analyzing how contribution rates increase with positive portfolio performance, while ξ could reflect reductions in contributions during market downturns. These parameters can also be tuned to match empirical risk aversion patterns observed in lifecycle funds.

1.3 How the Benchmark Affects and Works within Our Model

Our model studies optimal portfolio selection under **Hyperbolic Absolute Risk Aversion (HARA)** utility, explicitly considering **regret aversion** tied to this benchmark M_t.

HARA Utility Overview:

HARA utility is a flexible framework where risk aversion varies with wealth, making it ideal for modeling regret relative to a benchmark. It generalizes common utility functions (e.g., power or exponential utility) and allows analytical tractability in our continuous-time setting. The HARA utility function used here is defined in **Section 3.4 in the outline file.**

Wealth and Benchmark Dynamics:

We have two key state variables evolving dynamically:

- **Wealth Process (W_t)**:
  - Reflects the total value of the portfolio, evolving based on asset returns (risky and risk-free) and contributions/withdrawals.
- **Benchmark Process (M_t)**:
  - Reflects the cumulative net contributions dynamically adjusted according to investor psychology as modeled by the differential equation given above.

Visualizing the Dynamics:

To illustrate, consider a figure showing the evolution of wealth W_t and benchmark M_t over time. In regions where $W_t > M_t$, contributions increase due to optimism, while in regions where $W_t < M_t$, contributions decrease due to regret. [A figure could be inserted here as we start formally writing our paper, e.g., a line plot of W_t and M_t with annotated regions for optimism ($W_t > M_t$) and regret ($W_t < M_t$).]

Regret Aversion:

- **Regret** is defined as dissatisfaction triggered by wealth outcomes below the benchmark. Specifically, investors measure their success as:
  - Wealth W_t relative to cumulative contributions M_t.
- **Regret aversion** means investors have a strong preference to avoid falling behind this benchmark:
  - The closer wealth W_t is to M_t (especially if $W_t < M_t$), the stronger their regret aversion and thus the more risk-averse their behavior becomes.

1.4 Achieving Our Research Contributions through the Benchmark

Let's directly connect our benchmark modeling choice to each of our stated research contributions **Refer to the Introduction part-Main Contribution in the outline file**:

Contribution 1: Formulation of a Continuous-time Control Problem with Regret

- Our benchmark definition directly embeds regret aversion into our model.
- The difference $(W_t - M_t)$ explicitly appears in the investor's utility function:
  - Investors maximize expected utility defined over the gap $W_t - M_t$.
  - Formally, the investor solves:
$$\max_{\alpha_t} E\left[U(W_T - M_T)\right],$$
    subject to the wealth and benchmark dynamics we've established.
- Thus, the continuous-time control problem explicitly incorporates the **regret term**.

Contribution 2: Deriving Closed-form Expressions for Optimal Asset Shares under HARA with Regret Aversion

- With the benchmark explicitly defined as cumulative contributions, our utility function maintains **homotheticity**:
  - This allows for analytical tractability.
  - Specifically, we assume HARA utility:
$$U(W_t - M_t) = \frac{1}{1-\gamma}\left[a(W_t - M_t) + b\right]^{1-\gamma}, \quad a > 0, b \geq 0$$
- Because this benchmark process is deterministic conditional on wealth and contributions, the resulting **Hamilton–Jacobi–Bellman (HJB)** PDE simplifies significantly, enabling us to derive closed-form solutions or tractable approximations for the optimal fraction of wealth invested in risky assets.**I will derive this HJB and go over and compare the HJB that James has derived**

Contribution 3 (Core Contribution): Comparative Statics—Benchmark Proximity Amplifies Risk Aversion

- This benchmark definition is critical in clearly demonstrating how regret aversion shapes optimal investment:
  - When $W_t \approx M_t$, investors become acutely sensitive to deviations.
  - When $W_t < M_t$, regret intensifies, amplifying risk aversion substantially.
- Formally, comparative statics can be derived:

- Sensitivity parameters η, ξ directly influence how quickly investors adjust contributions, thus influencing wealth dynamics and the probability of entering "regret zones."
  - By explicitly quantifying how increased regret aversion (through parameter variation) could leads to lower risky asset allocation as wealth approaches the benchmark, thus capturing key behavioral patterns observed empirically.

- **Example for intuition**:
  - Suppose at time $t = 5$ years, cumulative contributions $M_5 = \$100,000$, with parameters $c_0 = 5000$, $\eta = 0.1$, $\xi = 0.2$.
  - If wealth $W_5 = \$95,000$ (below benchmark), the contribution rate adjusts as:

    $$dM_t = \left[5000 + 0.1(95000 - 100000)^+ - 0.2(100000 - 95000)^+\right] dt = [5000 - 0.2 \cdot 5000]\, dt = 4000\, dt,$$

    reducing contributions due to regret. Regret aversion heightens, and the investor allocates less to risky assets (e.g., reducing equity share from 60% to 50% to avoid further losses, assuming a market with mean return $\mu = 8\%$ and volatility $\sigma = 20\%$).
  - If $W_5 = \$105,000$, the contribution rate becomes:

    $$dM_t = \left[5000 + 0.1(105000 - 100000)^+ - 0.2(100000 - 105000)^+\right] dt = [5000 + 0.1 \cdot 5000]\, dt = 5500\, dt,$$

    increasing contributions due to optimism. The investor may increase equity share to 65%, reassured by positive performance.

Contribution 4: Implications for Pension Design and Lifecycle Investing

- This cumulative net contribution benchmark realistically models pension saving behaviors and lifecycle investment patterns.
- Explicitly highlights the role of benchmark-based regret:
  - Pension fund managers can use this insight to design contribution schedules or recommend asset allocations sensitive to investor psychological reactions.
  - Lifecycle investment products can adjust risk profiles dynamically as wealth approaches cumulative contributions.

Assumptions and Limitations

Our model assumes contributions adjust instantaneously based on the wealth-benchmark gap $W_t - M_t$, which may oversimplify real-world frictions like transaction costs or delayed decisions. Parameters η and ξ require empirical calibration, which can be challenging due to limited data on regret-driven behavior. The continuous-time framework ensures mathematical tractability but may differ from discrete-time applications in practice, where contributions occur at fixed intervals (e.g., monthly).